VARUNIKA NAINI

Hyderabad, Telangana varunikaanaini@gmail.com +91 7032599534 GitHub LinkedIn

EDUCATION

Indian Institute of Information Technology Design and Manufacturing, Kurnool *Nov. 2022 – Present*
BTech in Computer Science (GPA: 8.62)

PROFESSIONAL EXPERIENCE

Dheyo AI *May 2025 – Present*
AI & ML Intern

- Designed a 4-stage **sim-to-real synthetic data pipeline** for robotic grocery picking: **Gemini API** converts scene configs into physics-verified 3D placement JSONs encoding per-item location, quaternion rotation, face direction, and spatial constraints; a custom **3D OBB renderer** with OpenCV z-buffer and depth-encoded orientation maps feeds a fine-tuned **LoRA** model on **Qwen Image Edit** to synthesize photorealistic crate images; **Apple DepthPro** and **SAM3** then generate per-instance depth maps and segmentation masks with a **preferential** flag marking the shallowest object as the robot's priority pick target.
- Generated LoRA training corpora entirely in **Blender** – UV-textured per-face assets, physics drop simulations, metadata extraction, and photorealistic target synthesis; fine-tuned with **LoKr** and **LoRA** at bf16 precision, cutting manual image collection by around **60-70%** and enabling deterministic spatial control for downstream robot simulation.
- Built a GRPO-based prompt enhancement framework with **custom-designed reward signals**, raising output coherence scores by **70%** against untuned baseline prompts.
- Developed a pipeline that **dynamically determines the optimal quantization schema per layer**, outperforming Unsloth dynamic quantization by **2–4%** on benchmark scores; accelerated quantized LLM inference **11,000**× through **CUDA** and **Numba** kernel rewrites, benchmarked across 5 tasks with throughput parity to GGUF-based models.
- Explored **UR10** and **Franka Panda** robotic arms alongside the **GR1T2** humanoid platform in **Isaac Sim** – studying joint kinematics, motion planning, and pick-and-place task execution, and comparing parallel-jaw gripper versus humanoid manipulation for sim-to-real generalization.

Suvidha Foundation *Jun. 2024 – Jul. 2024*
Machine Learning Intern

- Benchmarked KL-Sum, T5, Pegasus, and BERT for abstractive text summarization; recorded ROUGE scores of **0.45** on DUC 2001 and **0.52** on CNN/Daily Mail.
- Lowered model error rates by **15%** via hyperparameter tuning and architectural ablation across all four summarization models.

Trainity *May 2024 – Jun. 2024*
Data Analyst Intern

- Completed **8** data analysis projects using Python, SQL, and Power BI – spanning data cleaning, transformation, and dashboard delivery.
- Identified patterns that improved client decision-making efficiency by **20%**, informing business strategy across multiple accounts.

TECHNICAL SKILLS

Languages: Python, R, SQL, C, C++, HTML, CSS, JavaScript

Data & Machine Learning: PyTorch, TensorFlow, Scikit-learn, CUDA, Numba, NLP, Power BI, Tableau, OpenCV, MediaPipe, LoRA, vLLM

Tools & Infra: PostgreSQL, MySQL, Git, Jupyter Notebook, Isaac Sim, Blender, OOP, Data Structures & Algorithms

PROJECT EXPERIENCE

LBA-Net: Boundary-Aware Self-Distillation for Efficient Semantic Segmentation

- A compact MobileNetV2 encoder-decoder with a dual-branch boundary-attentive decoder and EMA-based self-distillation, achieving **91.86% mIoU** on THRS-RSNA at only **13.88M parameters** – with **10**× fewer FLOPs than Swin-UNet while surpassing ARAA-Net, CaraNet, and Swin-UNet across **6 medical and natural-image benchmarks**.
- An EMA teacher-student scheme that stabilizes low-capacity encoder optimization sans an external teacher, delivering **+2.2% boundary IoU** over the unregularized baseline; cross-dataset evaluation spans THRS-RSNA, JSRT, Montgomery, Kvasir-SEG, PASCAL VOC, and ADE20K with no domain-specific retraining.

RDIF: Radiomic-Guided Diffusion Framework for Explainable Medical Segmentation

- A post-hoc XAI framework fusing **Integrated Gradient** seed CAMs with radiomic texture gates (Gabor, LBP, GLCM) via **Perona–Malik anisotropic diffusion**, yielding boundary-aligned saliency maps; RDIF CAM achieves **mIoU 0.556** on CVC-ClinicDB vs. **0.036** for LayerCAM – a **15**× gain – and near-perfect Pointing Game scores of **0.990** (THRS Epiphysis) and **0.986** (Montgomery).

- XAI-validated knowledge distillation confirms student activations converge to clinically relevant teacher patterns across **10 benchmarks** (X-ray, endoscopy, natural images); Pearson r of **0.687** on THRS Epiphysis vs. **0.044** (GradCAM) quantifies structural alignment between saliency and ground-truth anatomy.

Predictive Analysis for Customer Churn

- Predicted telecom customer churn at **89% accuracy** by ensembling Random Forest and XGBoost; applied SHAP to surface the top churn drivers for the client team.
- Reduced projected churn by **25%** through retention recommendations visualized in Matplotlib and Seaborn dashboards.

Sentiment Analysis on Social Media Data

- Fine-tuned BERT on Twitter customer reviews with TensorFlow, reaching an **F1-score of 0.92** – a **9-point gain** over a bag-of-words baseline.
- Packaged the inference code as a Python module to enable sentiment scoring in downstream analytics pipelines.

Gesture Game Controller

- Developed a gesture-controlled gaming interface with OpenCV and MediaPipe, cutting input latency by **30%** compared to keyboard controls.
- Integrated Arduino for hardware actuation, connecting vision-model output directly to physical game peripherals.

CERTIFICATIONS

Kaggle – Intro to Machine Learning *(Kaggle, 2024)* – Decision trees, model validation, and overfitting trade-offs using Scikit-learn on structured datasets.

Kaggle – Intermediate Machine Learning *(Kaggle, 2024)* – Pipelines, cross-validation, XGBoost, and data leakage prevention on tabular data.

Kaggle – Intro to Deep Learning *(Kaggle, 2024)* – Neural networks, dropout regularization, and batch normalization using Keras on benchmark datasets.